Physiology Lab2.
2nd Class
2026-2027

WHITE BLOOD CELL (WBC) COUNT (LEUKOCYTE COUNT)

Objectives

- Indicate the importance of doing total WBC count.
- Do the total WBC count by the manual method.
- Describe the normal total WBC count.
- Define leukocytosis, leukopoenia, and physiological and Pathological conditions in which they are seen.

Introduction:

- The white blood cells (WBCs; leukocytes) constitute the major defense system of the body against invasion by microbes, toxins and other foreign invaders. Their number is kept constant in health, but it increases or decreases in many diseases.
- A clinician generally wants this test done, along with differential count, Hb, etc. as part of "full blood count" (FBC or complete blood count).

Methods:

1. **Hemocytometry (manual method)**
2. Automatic Electronic Cell Counters

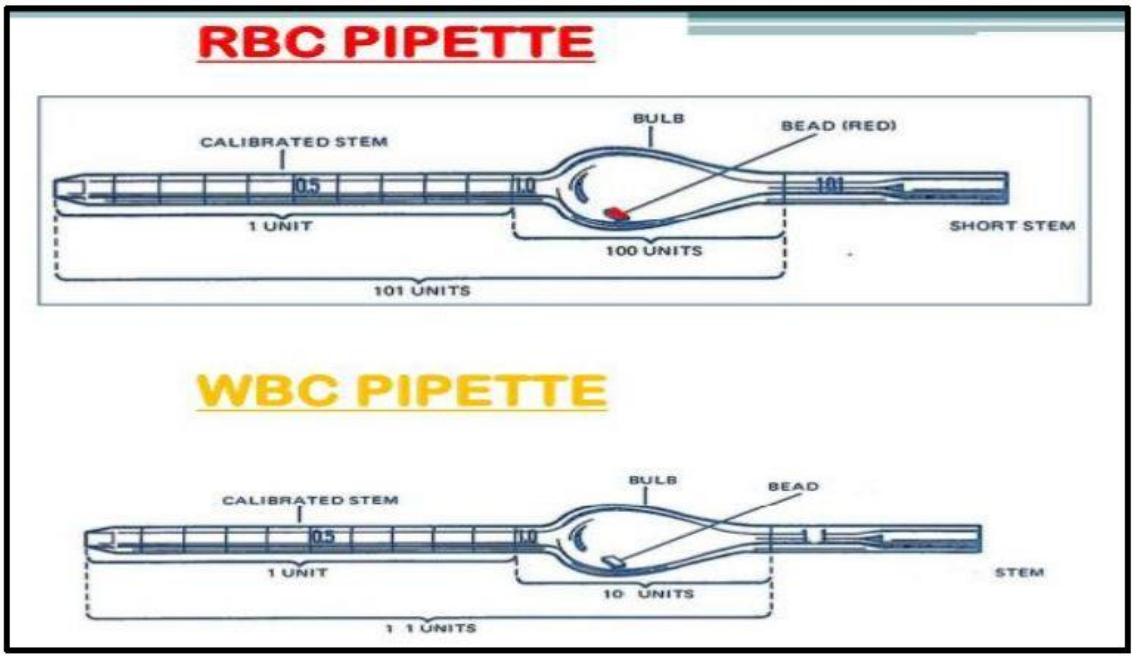
Principle:

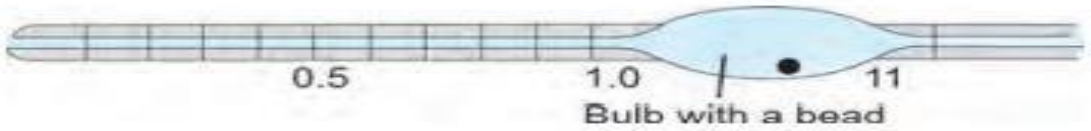
A sample of blood is diluted with a diluting fluid which destroys the red cells and stains the nuclei of the leukocytes. The cells are then counted in a counting chamber and their number in undiluted blood reported as leukocytes/mm

Materials:

- 1. Microscope •Blood lancet/pricking needle. •Sterile cotton/gauze swabs. •70% alcohol.
- 2. WBC pipettes: white bead in bulb, and markings 0.5, 1.0, and 11.

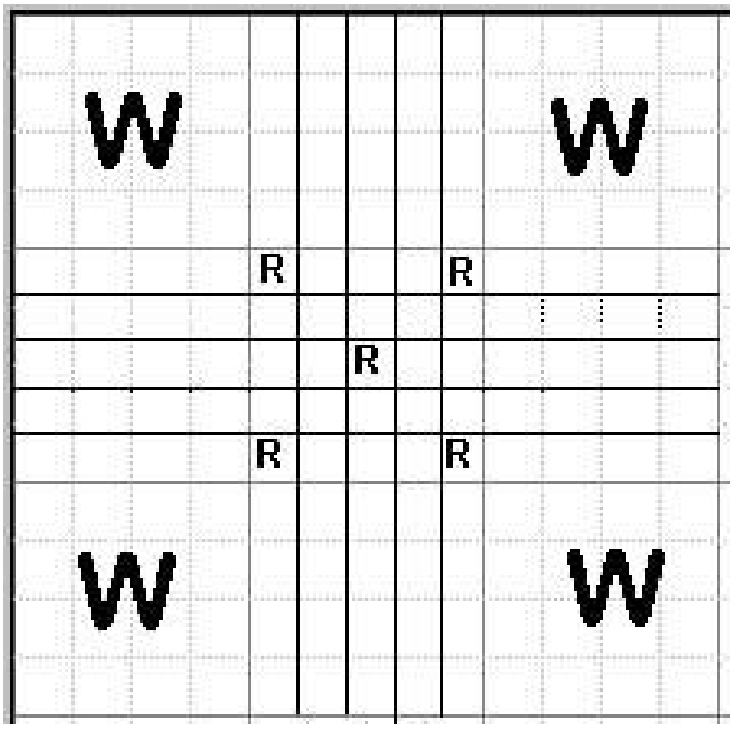
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• *Clean and dry pipettes, with free-rolling beads are required*.
• 3. **Turk's fluid**. This fluid is used for diluting the blood. It includes:
-Glacial acetic acid = 1.5 ml (hemolyzes RBCs without affecting WBCs).
-Gentian violet (1% solution) =1.5 ml (it stains the nuclei of leukocytes).
-Distilled water to 100 ml

•4.Counting chamber with coverslip.
• 4 WBC counting squares
• Each contains 16 squares



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*Procedure:

1. Obtain a drop of blood in the same manner as in RBC count. Draw blood up to the mark 0.5 using a WBC pipette.

2. Aspirate diluting fluid up to mark 11.

3. Remove blood from the outside of the pipette with a clean guaze.

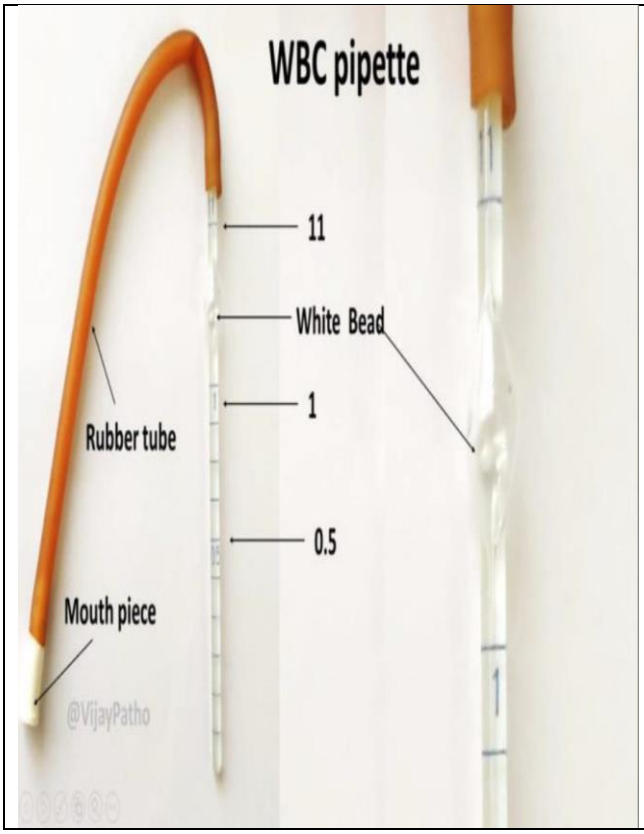
4. Gently rotate the pipette horizontally with your hand to ensure a proper amount of mixing for 3 minutes.

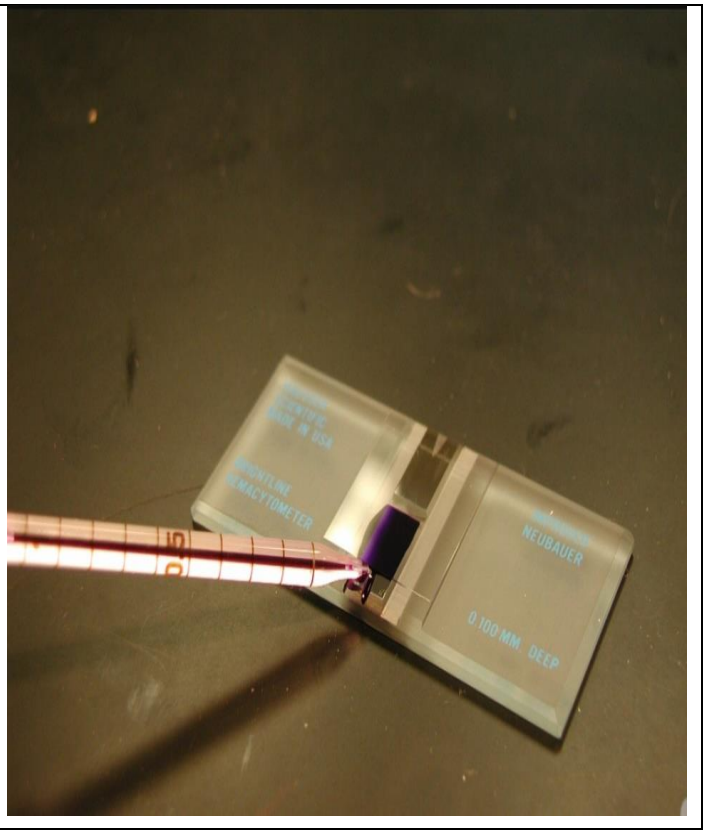
5. After mixing, discard the first four drops of the mixture.

6. Fill the counting chamber with diluted blood by holding the pipette at 45° with the surface of the slide and allow the mixture to spread under the coverslip. The filled chamber should be allowed to stand for about 1 minute before counting.

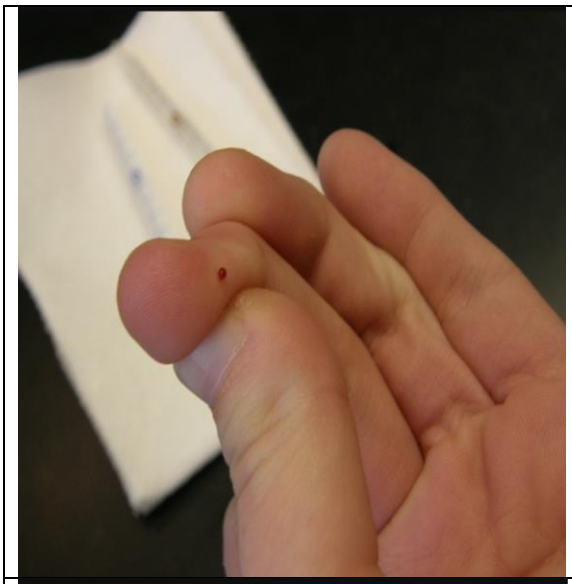
7. Count the WBC using low power 10x objective. Scan the four large corner squares marked A, B, C and D on the counting chamber.

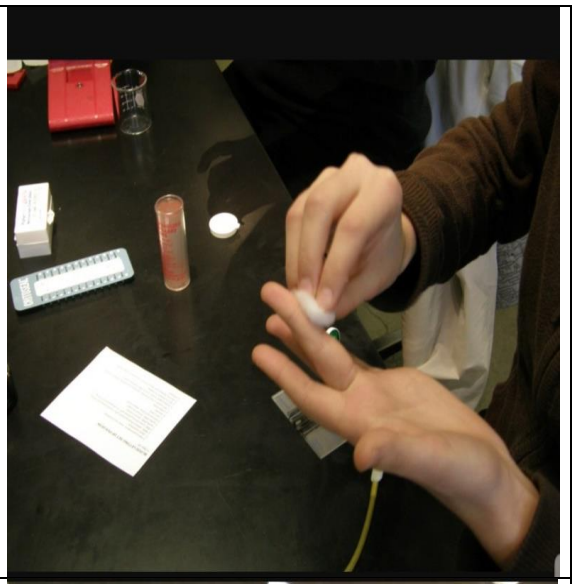
8. Count all WBCs in four large corner squares and add the results together to obtain the total number of cells counted. Calculation = X * 200.

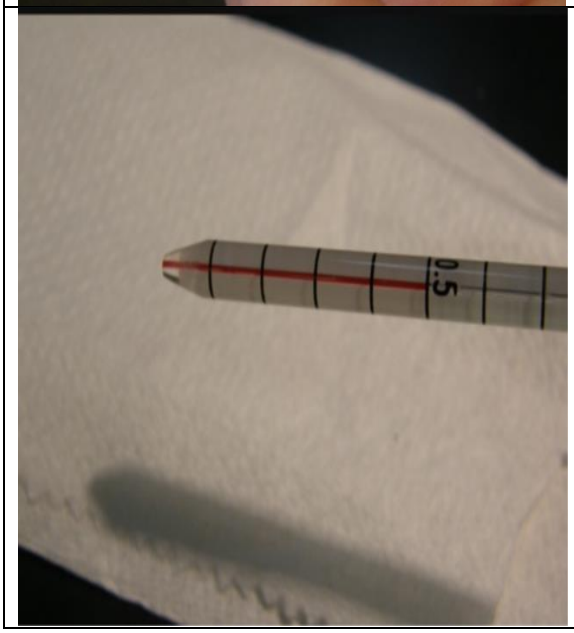


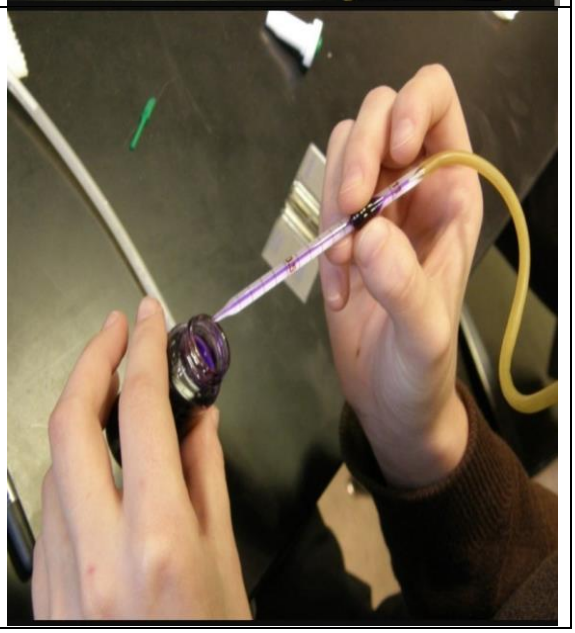


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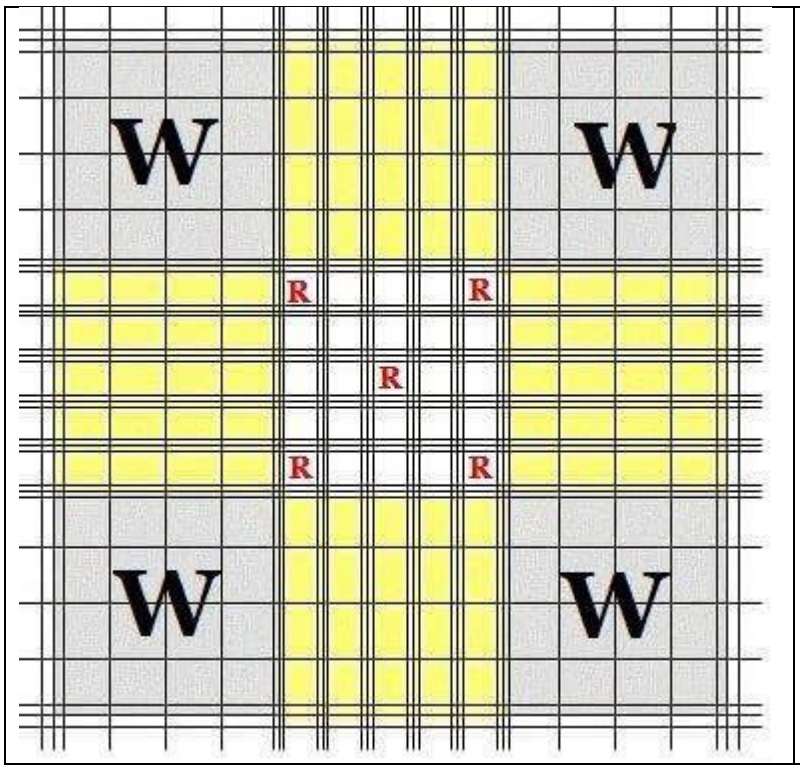


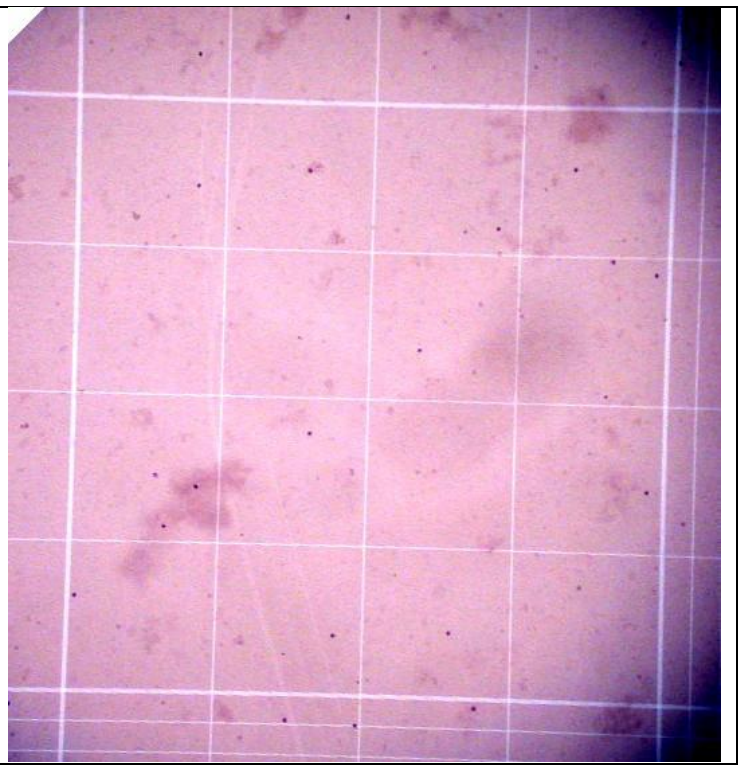






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Interpretation of the result:

• Normal value: Adult 4000-11000 cell/mm³

• Significance of the test :

Leukocytosis: The term refers to an increase in the number of WBCs beyond 11000/mm³. It is seen in the following conditions: - Physiological :normal infants, pregnancy, parturition& extremes of temperatures. -Pathological : Bacterial infection & Malignancies.

• **Leukopenia**: The term refers to a decrease in the number of white cells. - Physiological: rare - Pathological: Viral infections, drugs & repeated exposures to radiation